Telco Customer Churn Analysis

Project Documentation

Python | SQL Server | Excel | Power BI

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A beginner-level practice project

1. Introduction

Churn means a customer stops using a company's service. For a telecom company, every customer who leaves is lost monthly income, and getting a new customer costs more than keeping an old one.

In this practice project I analysed telecom customer data to understand how many customers leave, who they are, and why they may be leaving. I made it in March 2026 while I was learning the Data Analyst course at QSpiders. I had completed my BSc IT. I built it while following a YouTube tutorial and used the same dataset. I did the cleaning, analysis and dashboards myself using Python, SQL Server, Excel and Power BI.

2. Objectives

- Find the overall churn rate.
- Check how contract type, tenure, internet service and payment method affect churn.
- Compare monthly charges of churned and retained customers.
- Check if extra services like Online Security help customers stay.
- Build dashboards in Excel and Power BI.
- Write simple suggestions to reduce churn.

3. Dataset

The raw data file is telco_churn_unclean.csv. It has 7,048 rows and 21 columns. After cleaning, it has 6,910 rows and 23 columns (2 new columns were added).

Important columns

Column	Meaning
customerID	Unique ID of the customer
gender, SeniorCitizen, Partner, Dependents	Customer details
tenure	Number of months the customer has stayed
InternetService, OnlineSecurity, TechSupport, etc.	Services taken by the customer
Contract	Month-to-month, One year or Two year
PaymentMethod	How the customer pays the bill
MonthlyCharges, TotalCharges	Money charged to the customer
Churn	Yes if the customer left, No if the customer stayed
ChurnFlag (new)	1 for churned and 0 for retained
TenureGroup (new)	Tenure grouped as 0-12, 13-24, 25-48 and 49-72 months

4. Tools Used

Tool	Used for
Python (Pandas, Matplotlib)	Data cleaning and simple charts
Google Colab	Running the Python notebook
SQL Server (SSMS)	Loading data and running 10 queries
Microsoft Excel	Pivot tables, charts and dashboard
Power BI	Final dashboard

5. Data Cleaning in Python

The notebook Telco_Churn_Python_Analysis.ipynb does the cleaning with simple Pandas code. The steps are:

Step	What was done
1	Loaded the raw CSV file. Shape: 7,048 rows and 21 columns.
2	Checked data types and missing values.
3	Removed the $ sign and commas from MonthlyCharges and TotalCharges and converted them to numbers.
4	Removed rows with missing values and duplicate rows. Shape became 6,910 rows.
5	Changed SeniorCitizen from 0/1 to No/Yes (No: 5,791 and Yes: 1,119).
6	Created ChurnFlag (Yes = 1, No = 0).
7	Created TenureGroup using pd.cut.
8	Fixed the contract name 'month to month' to 'Month-to-month'.
9	Saved the cleaned file as telco_churn_cleaned_python.csv (6,910 rows and 23 columns).

6. Python Charts

Simple bar charts were made with Matplotlib and saved in the images folder. Three of them are shown below.

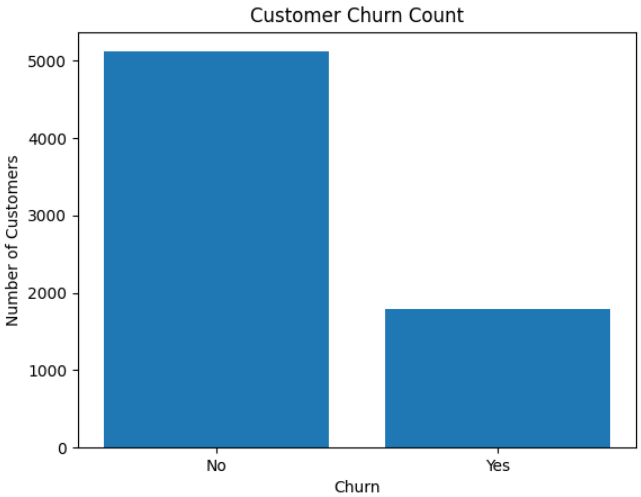


Figure 1: Customer churn count

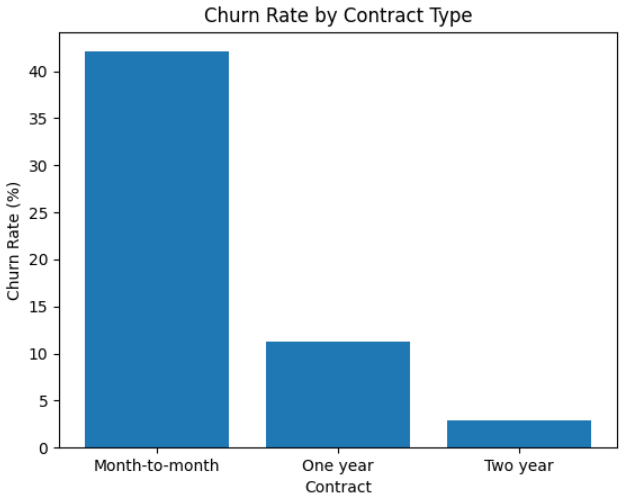


Figure 2: Churn rate by contract type

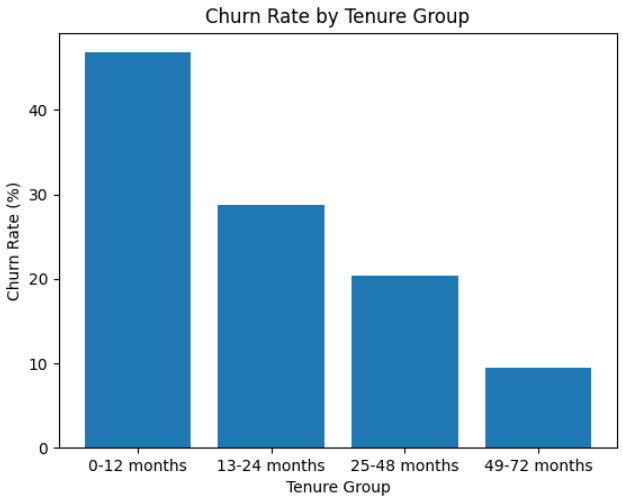


Figure 3: Churn rate by tenure group

7. SQL Analysis

The cleaned CSV was imported into SQL Server as the table telco_churn in the database telco_db. The file Telco_Churn_Analysis.sql has 10 queries. Screenshots of the results are in the sql folder.

One small fix was made in SQL with a single UPDATE: the contract value 'month to month' was changed to 'Month-to-month'.

Queries and results

Query	Purpose	Result
1	Check the table	6,910 rows
2	Overall churn rate	6,910 customers, 1,793 churned, 25.95%
3	Churn by contract	Month-to-month 42.07%, One year 11.28%, Two year 2.85%
4	Churn by internet service	Fiber optic 41.11%, DSL 18.45%, No internet 6.92%
5	Churn by payment method	Electronic check 44.43%, Mailed check 18.29%, Bank transfer 16.57%, Credit card 15.19%
6	Churn by tenure group	0-12 months 46.81% down to 49-72 months 9.51%
7	Churn by senior citizen	Yes 40.66%, No 23.10%
8	Churn by gender	Female 26.47%, Male 25.43%
9	Average monthly charges	Churned 74.97, Retained 61.53
10	Monthly revenue lost	134,423.60

8. Excel Dashboard

In Excel, the cleaned data was kept in a table named telco_data. Five pivot tables were made (Contract, Tenure, Retention, Security and Charges) and each one has a chart. All five charts are placed on one Dashboard sheet.



Figure 4: Excel dashboard

9. Power BI Dashboard

The cleaned CSV was loaded into Power BI. The dashboard has four KPI cards and several charts.

KPI card	Value
Total Customers	6,910
Churn Rate %	25.95%
Churned Customers	1,793
Revenue Loss	134.42K

- Retained Customers by Risk Type (Normal: 4,530 and High Risk: 587).
- Average of Monthly Charges by Churn.
- Count of Monthly Charges by Online Security.
- Churn Rate % by Tenure Band (0-6, 6-12 and 12+ months).
- Sum of Monthly Charges by Customer Type.
- Churned Customers by Contract.
- Customer Retention Overview (donut chart: 25.95% churned and 74.05% retained).

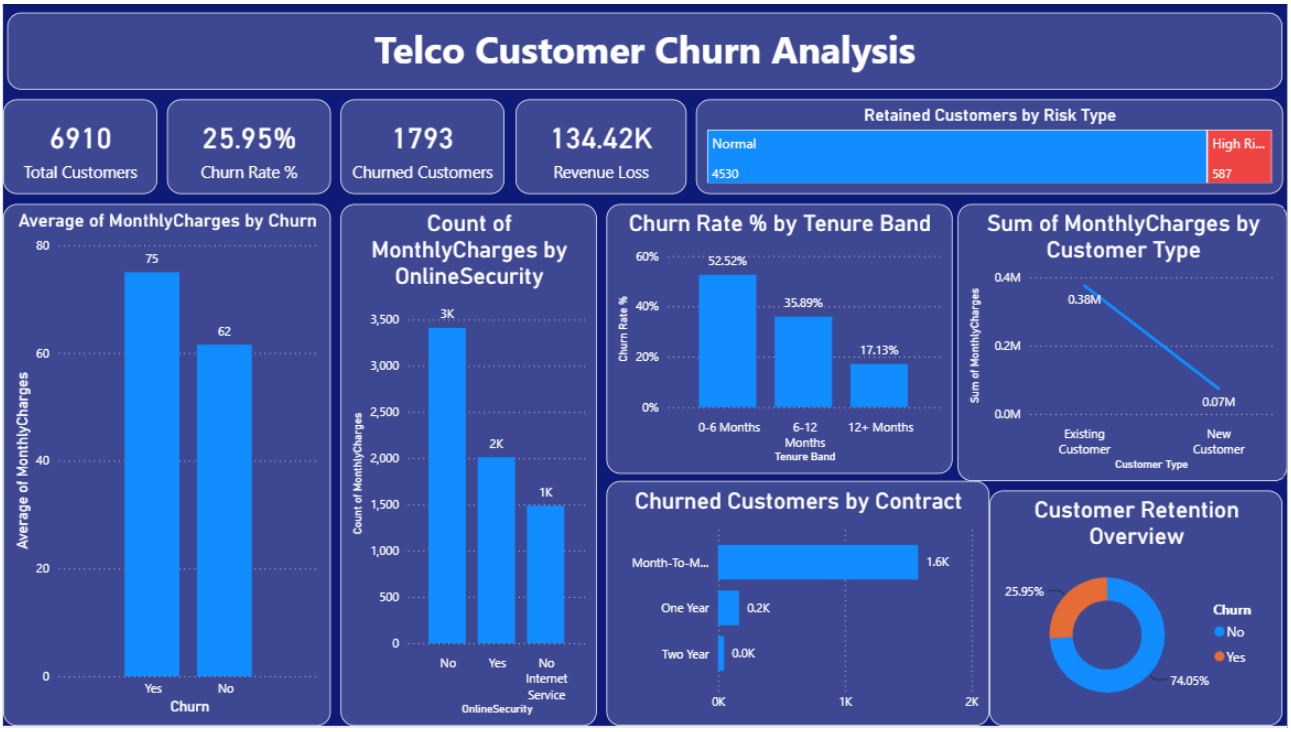


Figure 5: Power BI dashboard

Note: The Power BI dashboard uses 3 tenure bands (0-6, 6-12 and 12+ months), while Python, SQL and Excel use 4 tenure groups. Both show that new customers churn the most.

10. Key Findings

- The overall churn rate is **25.95%** (1,793 out of 6,910 customers).
- **Month-to-month** contracts have the highest churn (42.07%). Two year contracts have only 2.85%.
- New customers leave the most. Churn is 46.81% in the first 12 months and 9.51% after 49 months.
- **Fiber optic** customers churn at 41.11%, much more than DSL (18.45%).
- **Electronic check** users churn the most (44.43%). Automatic payment users are around 15% to 17%.
- Senior citizens churn more (40.66%) than other customers (23.10%).
- Customers without Online Security churn at 40.93%, but with Online Security only 14.62%.
- Churned customers pay more on average (74.97) than retained customers (61.53).
- Gender does not make a big difference (26.47% and 25.43%).
- About **134.42K** of monthly revenue is lost because of churn.

11. Suggestions

- Give offers to move month-to-month customers to one or two year contracts.
- Give extra support and offers to customers in their first year.
- Check the quality and price of the fiber optic service.
- Promote Online Security and Tech Support to customers who do not have them.
- Encourage customers to change from electronic check to automatic payment.
- Give special attention to senior citizens.

12. Limitations

- This is a practice project made while learning, so only simple methods are used.
- The analysis shows which groups churn more, but it does not prove the exact reason for churn.
- No machine learning model was built. This can be a future improvement.

13. Conclusion

This project helped me practice the full analytics process: cleaning data in Python, analysing it with SQL, building an Excel dashboard and a Power BI dashboard. The main result is that churn is highest among new customers on month-to-month contracts who use fiber optic internet and pay by electronic check. The company can reduce churn by moving these customers to longer contracts and giving them better support.

14. Project Files

Folder	Files
data	telco_churn_unclean.csv, telco_churn_cleaned_python.csv
python	Telco_Churn_Python_Analysis.ipynb
sql	Telco_Churn_Analysis.sql and result screenshots
excel	telco_churn_cleaned_python.xlsx, excel_dashboard.png
powerbi	Customer_Churn_Analysis.pbix, powerbi_dashboard.png
images	Python chart images